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DOA Estimation by Reduced-Dimension MUSIC with Uniform Circular Array

Meng Sun, Dongqi Shi, Jingjing Pan, Xiaofei Zhang, Yide Wang and H. C. So

***Abstract*—Uniform circular array (UCA) is widely employed in the field of direction-of-arrival (DOA) estimation, due to its uniform coverage in all directions and its ability to estimate both azimuth and elevation angles. Conventional methods, such as two-dimensional MUSIC (2D-MUSIC) or beam-space transform based methods, either require computationally intensive multi-dimensional searches, or suffer from residual error and bias in DOA estimation, which limit their practicality in real-time applications. To address these challenges, we devise a reduced-dimension MUSIC (RD-MUSIC) for DOA estimation by investigating both odd and even numbers of UCA sensors. The proposed method rearranges the sensor positions, and then splits the steering vector into two components, allowing for separate 1D searches for the elevation and azimuth angles. Our solution can achieve high estimation accuracy in a computationally attractive manner. The effectiveness of the RD-MUSIC is evaluated through numerical simulations as well as laboratory and field experiments, demonstrating its robustness at different signal-to-noise ratio levels and number of snapshots. The results highlight the potential of RD-MUSIC for real-time applications where computational resources are limited and high accuracy is required.**

***Index Terms*—Direction-of-Arrival (DOA) Estimation, Reduced-Dimension MUSIC, Uniform Circular Array (UCA).**

I. Introduction

Direction-of-arrival (DOA) estimation is a fundamental problem in array signal processing, with wide-ranging applications in radar, sonar, and wireless communications [1]–[13]. This is also important for smart vehicles, ranging from autonomous driving and collision avoidance, providing the spatial awareness that keeps vehicles safely in lane [14], [15]. The primary objective of DOA estimation is to determine the angles of incoming signals impinging on sensor array. However, the performance of DOA estimators is highly dependent on the array geometry, including the number of sensors, their spatial arrangement, and inter-element spacing. Therefore, choosing an appropriate array structure is crucial to enhance estimation accuracy, resolution, and robustness, especially in challenging environments with limited array aperture, low signal-to-noise ratio (SNR), and coherent signals.

In recent years, considerable research efforts have been devoted to improving the DOA estimation performance of various array configurations, such as uniform linear array (ULA), planar array, L-shaped array and parallel linear array [13], [16]–[24]. In [16], a modified orthogonal propagator method is proposed for finding DOAs with ULA that incorporates spline interpolation to reduce the influence of noise. In [20], the authors design a sparse planar array for wide-band underwater acoustical imaging using multitask Bayesian compressive sensing, which jointly optimizes sensor locations and frequency-dependent weights to match desired beam patterns. For L-shaped arrays, [21] develops a convex optimization-based two-dimensional (2D) DOA estimator to enhance both virtual aperture and virtual snapshots by exploiting the generalized conjugate symmetry property to attain higher estimation accuracy and source identifiability.

Despite the above-mentioned array configurations, uniform circular array (UCA) has attracted significant attention due to its unique geometric advantages. Unlike linear arrays, UCA provides consistent performance for signals arriving from any angle, making it particularly suitable for practical applications. In [25], the authors propose a low-complexity 2D DOA estimation scheme with cell searching for a mobile relay station equipped with a UCA in handover regions. This method can jointly estimate DOAs and cell IDs, and is robust against symbol timing offsets. In addition, it is capable of simultaneously estimating both azimuth and elevation angles. In [26], the authors derive virtual concentric circular arrays by mounting a single linear array on a rotating platform that enable high-accuracy 2D DOA estimation with enhanced degrees-of-freedom (DOFs) and reduced mutual-coupling sensitivity. However, UCA also introduces several inherent challenges, including increased computational complexity (for example, 2D-MUSIC, is required to estimate both azimuth and elevation angles via 2D search) as well as difficulties in array calibration and signal modeling [27].

Over the past few decades, a number of algorithms have been developed to address the above issues [26], [28]–[34]. The beamspace transform (BT) based DOA estimators, such as BT-MUSIC and BT-ESPRIT, have been proposed for UCA [28], [29], [31]. The BT employs phase mode excitation principle, which enables the mapping of UCA steering vectors into a Vandermonde structure. Afterwards, subspace methods are applied for DOA finding. Similar to BT, in the work of [30], a virtual signal subspace based method is applied for DOA estimation by virtually transforming the UCA into a ULA with a Vandermonde structure and constructing a stable virtual transformation matrix using singular value decomposition. This method enables accurate DOA estimation without peak searching. However, BT based DOA estimators require

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Method	Estimation Approach	Property	Main Limitation
2D-MUSIC [27]	2D search	Unbiased estimator	Heavy computational loads
BT-ESPRIT [28]	Rotational-invariance closed-form	Biased (beamspace)	Requires large number of sensors
UCA-root-MUSIC [29]	Polynomial rooting	Unbiased estimator	Parity restriction (only for odd M)
Virtual ULA [30]	Matrix-pencil on virtual ULA	Unbiased estimator	Virtual-array approximation
PE-DRC-MUSIC [26]	Synthetic UCA via rotation	Unbiased estimator	Array movement required

TABLE I
RELATED WORKS ON DOA ESTIMATION WITH UCA

a sufficient number of UCA sensors to eliminate aliasing in the beamspace manifold. Their performance is degraded when the UCA comprises only a small number of sensors [28]. Secondly, the performance of such DOA estimators is highly sensitive to the number of UCA sensors: UCA with an odd number of sensors tend to have lower bias than those with an even number [28].

To tackle the above-mentioned challenges, in this paper, we propose a reduced-dimension MUSIC (RD-MUSIC) for DOA finding with UCA sensors. Compared with existing approaches, our method transforms the multi-dimensional search into a series of 1D searches for both odd and even numbers of UCA sensors. Specifically, it first rearranges the sensor positions according to the number of UCA sensors. Then, it decomposes the steering vector into two parts, each containing only one parameter related to the DOAs of the incoming signals. Finally, the RD-MUSIC estimates the angle parameters by identifying the peaks in the 1D-MUSIC pseudo-spectrum. The superiority of the RD-MUSIC is demonstrated via numerical simulations as well as laboratory and field experiments. Compared to its competitors, the RD-MUSIC provides superior DOA estimation accuracy with reduced computational complexity, and thus can be applied in real-time applications. The results also show that the RD-MUSIC performs robustly across different SNR levels, numbers of snapshots as well as number of UCA sensors.

The contributions of this paper are summarized as follows:

- Compared with 2D-MUSIC, RD-MUSIC significantly reduces the computational load, while maintaining high DOA estimation accuracy by converting the 2D search into 1D searches.
- The RD-MUSIC delivers consistent and reliable DOA estimation performance across UCAs even with small number of sensors, regardless of whether the number is odd or even.
- The RD-MUSIC can be employed in real-world applications.

The rest of the paper is organized as follows. Section II introduces the data model of the received signals. Section III describes the proposed method for DOA estimation with odd and even numbers of UCA sensors. Simulation and experimental results evaluating the performance of the RD-MUSIC compared with its competitors are provided in Sections IV and V, respectively. Conclusions are drawn in Section VI.

Notations: The symbols $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^*$ denote the transpose, conjugate transpose, and complex conjugate operators, respectively. $\mathbb{E}\{\cdot\}$, $\det\{\cdot\}$, $\min\{\cdot\}$ represent the expectation, determinant, and minimum operators, respectively. Bold lowercase and uppercase characters denote vectors and matrices, respectively. The diag and anti-diag refer to diagonalization and anti-diagonalization operators, respectively. $\mathbf{I}_M$ and $\mathbf{J}_M$ represent the $M \times M$ identity and anti-identity matrices, respectively. The symbol $\oplus$ denotes the Hadamard product.

Acronym	Definition
AIC	Akaike Information Criterion
BT	Beamspace Transformation
CRB	Cramér–Rao Bound
DOA	Direction-of-Arrival
DOF	Degree-of-Freedom
EVD	Eigenvalue Decomposition
MDL	Minimum Description Length
MIMO	Multiple-Input Multiple-Output
MUSIC	MUltiple SIgnal Classification
RD-MUSIC	Reduced-Dimension MUSIC
RMSE	Root-Mean-Square Error
SNR	Signal-to-Noise Ratio
UCA	Uniform Circular Array
ULA	Uniform Linear Array

TABLE II
LIST OF ACRONYMS.

II. DATA MODEL

Consider K incoming signals impinge on a UCA with M elements, as depicted in Fig. 1. The received signals contain angle information (both elevation angle θ_k and azimuth angle ϕ_k, $k = 1, 2, \ldots, K$) of sources relative to the UCA. Let $\mathcal{M} = \{0, \ldots, M-1\}$ be the set of M sensors. The signal received by the mth sensor with $m \in \mathcal{M}$ can be expressed as

$$\begin{aligned} y_m(t) &= \sum_{k=1}^{K} s_k(t) \exp\left\{ j2\pi \frac{R}{\lambda_k} \sin\theta_k \cos\{\phi_k - \gamma_m\} \right\} + \omega_m(t) \\ &= \sum_{k=1}^{K} s_k(t) \exp\left\{ j2\pi \frac{R}{\lambda_k} \sin\theta_k \cos\phi_k \cos\gamma_m \right\} \\ &\exp\left\{ j2\pi \frac{R}{\lambda_k} \sin\theta_k \sin\phi_k \sin\gamma_m \right\} + \omega_m(t) \\ &= \sum_{k=1}^{K} s_k(t) \exp\{ju_k \cos\gamma_m + jv_k \sin\gamma_m\} + \omega_m(t) \end{aligned} \tag{1}$$

where $s_k(t)$ is the signal emitted from the kth source. R represents the radius of the UCA, λ is the wavelength of the incoming signal. $\gamma_m = 2\pi m/M$, $u_k = 2\pi \frac{R}{\lambda} \sin\theta_k \cos\phi_k$ and

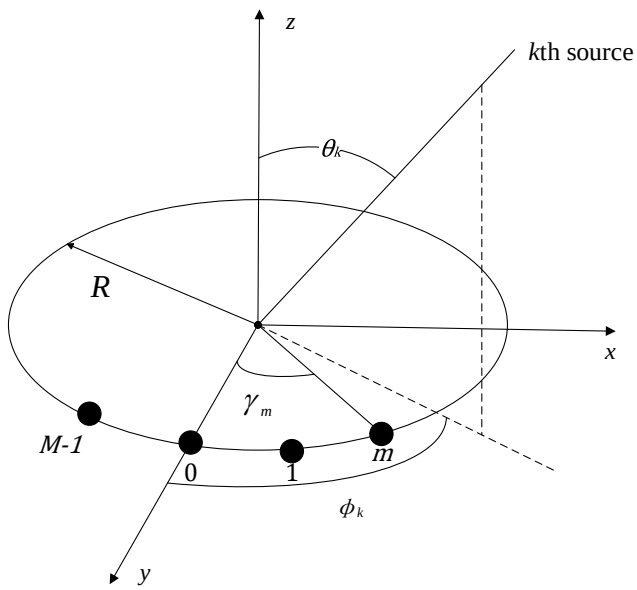


Fig. 1. UCA configuration.

$v_k = 2\pi\frac{R}{\lambda}\sin\theta_k \sin\phi_k$, $k = 1, \ldots, K$. $\omega_m(t)$ is assumed to be a zero-mean white Gaussian noise with variance σ^2. We can develop the following relationship:

$$\begin{aligned} u_k^2 + v_k^2 &= 4\pi^2 \frac{R^2}{\lambda^2} \sin\theta_k{}^2 \\ \frac{v_k}{u_k} &= \tan\phi_k. \end{aligned} \tag{2}$$

Note that when $\theta = 0$, the incoming signal impinges perpendicularly to the array plane, thus the phase is identical at every sensor:

$$\exp\{j2\pi R/\lambda \cdot \sin 0 \cdot \cos(\phi - \gamma_m)\} = 1 \quad \forall m.$$

The steering vector therefore becomes a constant and the array response no longer varies with the azimuth angle ϕ; any value of ϕ yields the same output. Consequently, ϕ is not identifiable at $\theta = 0$, which is an inherent "blind spot" common to all planar arrays at vertical incidence. The received signal can be grouped in the following vector form

$$\boldsymbol{y}(t) = \mathbf{A}\boldsymbol{s}(t) + \boldsymbol{\omega}(t) \tag{3}$$

where

- $\boldsymbol{y}(t) = [y_0(t), \ldots, y_{M-1}(t)]^T$ is the $M \times 1$ vector of the received signal;
- $\boldsymbol{s}(t) = [s_1(t), \ldots, s_K(t)]^T$ is the $K \times 1$ signal vector from K sources;
- $\boldsymbol{\omega}(t) = [\omega_0(t), \ldots, \omega_{M-1}(t)]^T$ is the $M \times 1$ noise vector with covariance matrix $\sigma^2\mathbf{I}_M$;
- $\mathbf{A} = [\boldsymbol{a}(u_1, v_1), \ldots, \boldsymbol{a}(u_K, v_K)]$ is the $M \times K$ steering matrix;
- $\boldsymbol{a}(u_k, v_k) = [\exp\{ju_k \cos\gamma_0 + jv_k \sin\gamma_0\}, \ldots, \exp\{ju_k \cos\gamma_{M-1} + jv_k \sin\gamma_{M-1}\}]^T$ is the $M \times 1$ steering vector.

The covariance matrix of $\boldsymbol{y}(t)$ is

$$\mathbf{Y} = \mathbb{E}\{\boldsymbol{y}(t)\boldsymbol{y}^H(t)\} = \mathbf{A}\boldsymbol{\Lambda}\mathbf{A}^H + \sigma^2\mathbf{I}_M \tag{4}$$

where $\boldsymbol{\Lambda} = \mathbb{E}\{\boldsymbol{s}(t)\boldsymbol{s}^H(t)\}$ is the covariance matrix of the incoming signals and $\mathbf{I}_M = [\boldsymbol{e}_0, \boldsymbol{e}_1, \ldots, \boldsymbol{e}_{M-1}]$ is the $M \times M$ identity matrix. As the signals come from different sources, they are uncorrelated. We apply eigenvalue decomposition (EVD) on $\mathbf{Y}$ to yield:

$$\mathbf{Y} = \boldsymbol{\Sigma}_s\boldsymbol{\Xi}_s\boldsymbol{\Sigma}_s^H + \boldsymbol{\Sigma}_n\boldsymbol{\Xi}_n\boldsymbol{\Sigma}_n^H \tag{5}$$

where $\boldsymbol{\Xi}_s$ and $\boldsymbol{\Xi}_n$ are diagonal matrices. The diagonal elements of $\boldsymbol{\Xi}_s$ are the K largest eigenvalues, whose associated eigenvectors are contained in $\boldsymbol{\Sigma}_s$; the diagonal elements of $\boldsymbol{\Xi}_n$ correspond to the remaining eigenvalues and the columns of $\boldsymbol{\Sigma}_n$ generate the noise subspace.

The number of sources K is either assumed to be known or accurately estimated from the data. For the latter case, we can use the classical Akaike information criterion (AIC) or minimum description length (MDL), which operates on the eigenvalues of the sample covariance matrix for source enumeration. [35]. Then, K is determined by searching the minimum value of the following AIC or MDL criterion:

$$K = \min_m [C(m)] \tag{6}$$

where $C(m)$ $(m = 0, 1, ...M-1)$ is expressed as:

$$C(m) = \begin{cases} -2N_t(M-m)\log\left\{\dfrac{\prod_{j=m+1}^{M}\lambda_j^{1/(M-m)}}{1/(M-m)\sum_{j=m+1}^{M}\lambda_j}\right\} & \text{AIC} \\ +2m(2M-m) & \\ -N_t(M-m)\log\left\{\dfrac{\prod_{j=m+1}^{M}\lambda_j^{1/(M-m)}}{1/(M-m)\sum_{j=m+1}^{M}\lambda_j}\right\} & \text{MDL} \\ +0.5m(2M-m)\log N_t & \end{cases}$$

Here, λ_j is the jth eigenvalue of the data covariance matrix (arranged in decreasing order), $j = 1, 2, ..., M$ while N_t is the number of snapshots.

The 2D-MUSIC can be applied to estimate both the elevation and azimuth angles, namely, θ_k and ϕ_k:

$$\begin{aligned} &\mathcal{P}_{\text{MUSIC}}(\theta, \phi) \\ &= \frac{1}{\boldsymbol{a}^H\{u(\theta,\phi),\ v(\theta,\phi)\}\boldsymbol{\Sigma}_n\boldsymbol{\Sigma}_n^H\boldsymbol{a}\{u(\theta,\phi),\ v(\theta,\phi)\}}. \end{aligned} \tag{7}$$

That is, the DOA estimates $\hat{\theta}_k$ and $\hat{\phi}_k$ are obtained by searching the peaks of (7). However, (7) requires 2D search, which is computational inefficient.

III. Proposed method

To avoid 2D search as in the 2D-MUSIC, RD-MUSIC is suggested in this work. Note that in the steering vector $\boldsymbol{a}(u, v)$, there are two terms: $\exp\{ju_k \cos\gamma_m\}$ and $\exp\{jv_k \sin\gamma_m\}$. To reduce the complexity, we propose to split the steering vector such that only 1D search is involved. To reduce the impact of noise, RD-MUSIC operates on the signal subspace $\boldsymbol{\Gamma} = \boldsymbol{\Sigma}_s\boldsymbol{\Sigma}_s^H$ instead of the full data covariance matrix, since the signal subspace contains all the information of the signals. Here, two cases are considered: M is odd and M is even.

A. UCA with Odd Sensor Number

When M is an odd number, we rearrange the sensor position numbers as $\mathcal{M}_1 = \{1,\ 2, \ldots, (M-1)/2,\ 0, (M+1)/2, \ldots M-1\}$. The steering vector can then be split as follows:

$$\begin{aligned}\tilde{\boldsymbol{a}}(u,\ v) = [&\exp\{ju\cos\gamma_1 + jv\sin\gamma_1\}, \ldots, \\ &\exp\{ju\cos\gamma_{M-1} + jv\sin\gamma_{M-1}\}]^T\end{aligned} \tag{8}$$

where $\tilde{\boldsymbol{a}}(u,\ v) = \mathbf{J}_o\boldsymbol{a}(u,\ v)$, with $\mathbf{J}_o = [\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{(M-1)/2}, \boldsymbol{e}_0, \boldsymbol{e}_{(M+1)/2}, \ldots, \boldsymbol{e}_{M-1}]$ being the permutation matrix. The following relationship between γ_m and γ_{M-m} are obtained: $\gamma_m + \gamma_{M-m} = 2\pi$, $\cos\gamma_m = \cos\gamma_{M-m}$ and $\sin\gamma_m = -\sin\gamma_{M-m}$. Then, $\tilde{\boldsymbol{a}}(u,\ v)$ can be rewritten as follows:

$$\begin{aligned}\tilde{\boldsymbol{a}}(u,\ v) &= \begin{bmatrix} e^{ju\cos\gamma_1 + jv\sin\gamma_1} \\ e^{ju\cos\gamma_2 + jv\sin\gamma_2} \\ \vdots \\ e^{ju\cos\gamma_0 + jv\sin\gamma_0} \\ \vdots \\ e^{ju\cos\gamma_{M-2} + jv\sin\gamma_{M-2}} \\ e^{ju\cos\gamma_{M-1} + jv\sin\gamma_{M-1}} \end{bmatrix} \\ &= \begin{bmatrix} e^{ju\cos\gamma_1} e^{jv\sin\gamma_1} \\ e^{ju\cos\gamma_2} e^{jv\sin\gamma_2} \\ \vdots \\ e^{ju\cos\gamma_0} e^{jv\sin\gamma_0} \\ \vdots \\ e^{ju\cos\gamma_2} e^{-jv\sin\gamma_2} \\ e^{ju\cos\gamma_1} e^{-jv\sin\gamma_1} \end{bmatrix} \\ &= \begin{bmatrix} e^{jv\sin\gamma_1} & & & & \\ & \ddots & & & \\ & & e^{jv\sin\gamma_0} & & \\ & & & \ddots & \\ & & & & e^{jv\sin\gamma_{M-1}} \end{bmatrix} \begin{bmatrix} e^{ju\cos\gamma_1} \\ e^{ju\cos\gamma_2} \\ \vdots \\ e^{ju\cos\gamma_0} \\ \vdots \\ e^{ju\cos\gamma_{M-2}} \\ e^{ju\cos\gamma_{M-1}} \end{bmatrix} \\ &= \begin{bmatrix} e^{jv\sin\gamma_1} & & & & \\ & \ddots & & & \\ & & e^{jv\sin\gamma_0} & & \\ & & & \ddots & \\ & & & & e^{jv\sin\gamma_{M-1}} \end{bmatrix} \begin{bmatrix} e^{ju\cos\gamma_1} \\ e^{ju\cos\gamma_2} \\ \vdots \\ e^{ju} \\ \vdots \\ e^{ju\cos\gamma_2} \\ e^{ju\cos\gamma_1} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} e^{jv\sin\gamma_1} & & & & & \\ & e^{jv\sin\gamma_2} & & & & \\ & & \ddots & & & \\ & & & 1 & & \\ & & & & \iddots & \\ & & & & e^{-jv\sin\gamma_2} & \\ & & & & & e^{-jv\sin\gamma_1} \end{bmatrix}}_{\mathbf{Q}(v)} \underbrace{\begin{bmatrix} e^{ju\cos\gamma_1} \\ e^{ju\cos\gamma_2} \\ \vdots \\ e^{ju} \end{bmatrix}}_{\boldsymbol{p}(u)}\end{aligned}$$

$\mathbf{Q}(v)$ and $\boldsymbol{p}(u)$ only contain v and u, respectively. The pseudo-spectrum of MUSIC can be expressed as

$$\begin{aligned}\mathcal{P}_{\text{MUSIC}}(u,\ v) &= \frac{1}{\tilde{\boldsymbol{a}}^H(u,\ v)\boldsymbol{\Sigma}_o\boldsymbol{\Sigma}_o^H\tilde{\boldsymbol{a}}(u,\ v)} \\ &= \frac{1}{\boldsymbol{p}^H(u)\underbrace{\mathbf{Q}^H(v)\boldsymbol{\Sigma}_o\boldsymbol{\Sigma}_o^H\mathbf{Q}(v)}_{\boldsymbol{\Psi}(v)}\boldsymbol{p}(u)}\end{aligned} \tag{9}$$

where $\boldsymbol{\Sigma}_o = \mathbf{J}_o\boldsymbol{\Sigma}_{RD}$, with $\boldsymbol{\Sigma}_{RD}$ being the noise subspace of $\boldsymbol{\Gamma}$. According to the Rayleigh quotient theorem, the quadratic form in (9) is equal to the minimum eigenvalue $\lambda_{\min}(v)$ of $\boldsymbol{\Psi}(v)$ [36], and thus (9) becomes:

$$\mathcal{P}_{\text{MUSIC}}(v) = \frac{1}{\lambda_{\min}(v)}. \tag{10}$$

Then, the K estimates of v_k, denoted by $\hat{v}_k$, $k = 1, \ldots, K$, are obtained by 1D peak search. With each of these K estimates, (10) can be written as a function of u:

$$\mathcal{P}_{\text{MUSIC}}(u) = \frac{1}{\boldsymbol{p}^H(u)\boldsymbol{\Psi}(\hat{v}_k)\boldsymbol{p}(u)}. \tag{11}$$

Then, u_k is estimated by searching the peak of (11) for each $\hat{v}_k$ $(k = 1, \ldots, K)$. As a result, a total of $(K+1)$ 1D searches are required. Employing the estimated values of u_k and v_k, and (2), the elevation and azimuth angles can be calculated as

$$\begin{aligned}\hat{\theta}_k &= \arcsin\left\{\frac{\lambda\sqrt{\hat{v}_k^2 + \hat{u}_k^2}}{2\pi R}\right\} \\ \hat{\phi}_k &= \arctan\left\{\frac{\hat{v}_k}{\hat{u}_k}\right\}\end{aligned} \tag{12}$$

B. UCA with Even Sensor Number

If M is an even number, the steering vector $\boldsymbol{a}$ in (3) can be written as

$$\begin{aligned}\boldsymbol{a}(u, v) &= \begin{bmatrix} e^{ju\cos\gamma_0 + jv\sin\gamma_0} \\ e^{ju\cos\gamma_1 + jv\sin\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M-2} + jv\sin\gamma_{M-2}} \\ e^{ju\cos\gamma_{M-1} + jv\sin\gamma_{M-1}} \end{bmatrix} = \begin{bmatrix} e^{ju\cos\gamma_0} e^{jv\sin\gamma_0} \\ e^{ju\cos\gamma_1} e^{jv\sin\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M-2}} e^{jv\sin\gamma_{M-2}} \\ e^{ju\cos\gamma_{M-1}} e^{jv\sin\gamma_{M-1}} \end{bmatrix} \\ &= \begin{bmatrix} e^{ju\cos\gamma_0} \\ e^{ju\cos\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M-2}} \\ e^{ju\cos\gamma_{M-1}} \end{bmatrix} \oplus \begin{bmatrix} e^{jv\sin\gamma_0} \\ e^{jv\sin\gamma_1} \\ \vdots \\ e^{jv\sin\gamma_{M-2}} \\ e^{jv\sin\gamma_{M-1}} \end{bmatrix} \\ &= \begin{bmatrix} e^{ju} \\ e^{ju\cos\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M/2-1}} \\ e^{-ju} \\ e^{ju\cos\gamma_{M/2-1}} \\ \vdots \\ e^{ju\cos\gamma_1} \end{bmatrix} \oplus \begin{bmatrix} 1 \\ e^{jv\sin\gamma_1} \\ \vdots \\ e^{jv\sin\gamma_{M/2-1}} \\ 1 \\ e^{-jv\sin\gamma_{M/2-1}} \\ \vdots \\ e^{-jv\sin\gamma_1} \end{bmatrix}.\end{aligned} \tag{13}$$

Since both $\sin\gamma_0$ and $\sin\gamma_{M/2}$ are equal to 0, $\exp\{jv_k\sin\gamma_0\}$ and $\exp\{jv_k\sin\gamma_{M/2}\}$ become 1 which are not functions

of v_k. Therefore, the 0th and $M/2$th sensor outputs are not employed, (13) is then modified as

$$\check{\boldsymbol{a}}(u,\ v) = \begin{bmatrix} e^{ju\cos\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M/2-1}} \\ e^{ju\cos\gamma_{M/2-1}} \\ \vdots \\ e^{ju\cos\gamma_1} \end{bmatrix} \oplus \begin{bmatrix} e^{jv\sin\gamma_1} \\ \vdots \\ e^{jv\sin\gamma_{M/2-1}} \\ e^{-jv\sin\gamma_{M/2-1}} \\ \vdots \\ e^{-jv\sin\gamma_1} \end{bmatrix}$$

$$= \begin{bmatrix} e^{ju\cos\gamma_1}e^{jv\sin\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M/2-1}}e^{jv\sin\gamma_{M/2-1}} \\ e^{ju\cos\gamma_{M/2-1}}e^{-jv\sin\gamma_{M/2-1}} \\ \vdots \\ e^{ju\cos\gamma_1}e^{-jv\sin\gamma_1} \end{bmatrix}$$

$$= \mathrm{diag}\{e^{jv\sin\gamma_1},\ldots,e^{jv\sin\gamma_{M/2-1}},\ e^{-jv\sin\gamma_{M/2-1}},\ldots,e^{-jv\sin\gamma_1}\}$$

$$\begin{bmatrix} e^{ju\cos\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M/2-1}} \\ e^{ju\cos\gamma_{M/2-1}} \\ \vdots \\ e^{ju\cos\gamma_1} \end{bmatrix} = \underbrace{\begin{bmatrix} e^{jv\sin\gamma_1} & & & & \\ & \ddots & & & \\ & & e^{jv\sin\gamma_{M/2-1}} & & \\ & & & e^{-jv\sin\gamma_{M/2-1}} & \\ & & & & \ddots \\ e^{-jv\sin\gamma_1} & & & & \end{bmatrix}}_{\tilde{\mathbf{Q}}(v)} \underbrace{\begin{bmatrix} e^{ju\cos\gamma_1} \\ \vdots \\ e^{ju\cos\gamma_{M/2-1}} \end{bmatrix}}_{\tilde{\mathbf{p}}(u)}$$

where $\check{\boldsymbol{a}}(u,\ v) = \mathbf{J}_e\boldsymbol{a}(u,\ v)$, with $\mathbf{J}_e = [\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{M/2-1}, \boldsymbol{e}_{M/2+1}, \ldots, \boldsymbol{e}_{M-1}]^T$ being the permutation matrix. We see that the steering vector $\check{\boldsymbol{a}}$ can also be decomposed as the product of $\tilde{\mathbf{Q}}(v)$ and $\tilde{\boldsymbol{p}}(u)$. Therefore, the MUSIC pseudo-spectrum with even number of UCA sensors can be expressed as

$$\begin{aligned} \tilde{\mathcal{P}}_{\text{MUSIC}}(u,\ v) &= \frac{1}{\check{\boldsymbol{a}}^H(u,\ v)\boldsymbol{\Sigma}_e\boldsymbol{\Sigma}_e^H\check{\boldsymbol{a}}(u,\ v)} \\ &= \frac{1}{\tilde{\boldsymbol{p}}^H(u)\underbrace{\tilde{\mathbf{Q}}^H(v)\boldsymbol{\Sigma}_e\boldsymbol{\Sigma}_e^H\tilde{\mathbf{Q}}(v)}_{\tilde{\boldsymbol{\Psi}}(v)}\tilde{\boldsymbol{p}}(u)} \end{aligned} \tag{14}$$

Here, the columns of $\boldsymbol{\Sigma}_e = \mathbf{J}_e\boldsymbol{\Sigma}_{RD}$ generate the noise subspace of $\boldsymbol{\Gamma}$ with even number of UCA sensors. Similar to the development of the odd number case, v can be estimated by searching the K peaks of the minimum eigenvalue $\tilde{\lambda}_{\text{min}}(v)$ of $\tilde{\boldsymbol{\Psi}}(v)$:

$$\tilde{\mathcal{P}}_{\text{MUSIC}}(v) = \frac{1}{\tilde{\lambda}_{\text{min}}(v)} \tag{15}$$

In the estimation of u, we need to separate two cases for consideration, namely, $M/2$ is odd and $M/2$ is even. For the former, $\tilde{\boldsymbol{p}}(u)$ in (14) can be decomposed as

$$\tilde{\boldsymbol{p}}(u) = [\tilde{\boldsymbol{p}}_1(u),\ \tilde{\boldsymbol{p}}_2(u),\ \tilde{\boldsymbol{p}}_3(u),\ \tilde{\boldsymbol{p}}_4(u)]^T \tag{16}$$

with

$$\begin{aligned} \tilde{\boldsymbol{p}}_1(u) &= [e^{ju\cos\gamma_1},\ldots,e^{ju\cos\gamma_{(M+2)/4-1}}]^T, \\ \tilde{\boldsymbol{p}}_2(u) &= [e^{ju\cos\gamma_{(M+2)/4}},\ldots,e^{ju\cos\gamma_{M/2-1}}]^T, \\ \tilde{\boldsymbol{p}}_3(u) &= [e^{ju\cos\gamma_{M/2+1}},\ldots,e^{ju\cos\gamma_{(3M-2)/4}}]^T, \\ \tilde{\boldsymbol{p}}_4(u) &= [e^{ju\cos\gamma_{(3M-2)/4+1}},\ldots,e^{ju\cos\gamma_{M-1}}]^T. \end{aligned}$$

They are related by

$$\tilde{\boldsymbol{p}}_1(u) = \tilde{\boldsymbol{p}}_3^*(u) = \mathbf{J}\tilde{\boldsymbol{p}}_2^*(u) = \mathbf{J}\tilde{\boldsymbol{p}}_4(u). \tag{17}$$

In order to construct a matrix $\tilde{\mathbf{P}}(u)$ with row conjugate symmetry, we swap $\tilde{\boldsymbol{p}}_2(u)$ and $\tilde{\boldsymbol{p}}_4(u)$ to yield:

$$\begin{aligned} \tilde{\mathbf{P}}(u) &= \begin{bmatrix} \mathrm{diag}\{\tilde{\boldsymbol{p}}_1(u)\} & \\ & \mathrm{diag}\{\tilde{\boldsymbol{p}}_4(u)\} \\ & \text{anti-diag}\{\tilde{\boldsymbol{p}}_3(u)\} \\ \text{anti-diag}\{\tilde{\boldsymbol{p}}_2(u)\} & \end{bmatrix} \\ &= \begin{bmatrix} \mathrm{diag}\{\tilde{\boldsymbol{p}}_1(u)\} & \\ & \mathrm{diag}\{\mathbf{J}\tilde{\boldsymbol{p}}_1(u)\} \\ & \text{anti-diag}\{\tilde{\boldsymbol{p}}_1^*(u)\} \\ \text{anti-diag}\{\mathbf{J}\tilde{\boldsymbol{p}}_1^*(u)\} & \end{bmatrix}. \end{aligned} \tag{18}$$

The steering vector $\check{\boldsymbol{a}}(u,\ v)$ in (14) can then be factorized as

$$\check{\boldsymbol{a}}(u,\ v) = \tilde{\mathbf{P}}(u)\tilde{\boldsymbol{q}}(v) \tag{19}$$

where $\tilde{\boldsymbol{q}}(v)$ is a vector parameterized by v only. The MUSIC pseudo-spectrum in (14) is then

$$\tilde{\mathcal{P}}_{\text{MUSIC}}(v,u) = \frac{1}{\tilde{\boldsymbol{q}}^H(v)\underbrace{\tilde{\mathbf{P}}^H(u)\boldsymbol{\Sigma}_e\boldsymbol{\Sigma}_e^H\tilde{\mathbf{P}}(u)}_{\tilde{\boldsymbol{\Phi}}(u)}\tilde{\boldsymbol{q}}(v)}. \tag{20}$$

For each estimate of v, i.e., $\hat{v}_k$, $k = 1,\cdots,K$, the corresponding u_k can then be estimated by searching the peak of the minimum eigenvalue $\tilde{\lambda}_{\text{min}}(u)$ of $\tilde{\boldsymbol{\Phi}}(u)$:

$$\tilde{\mathcal{P}}_{\text{MUSIC}}(u) = \frac{1}{\tilde{\lambda}_{\text{min}}(u)} \tag{21}$$

When $M/2$ is even, $\sin\gamma_0 = \sin\gamma_{M/2} = \cos\gamma_{M/4} = \cos\gamma_{3M/4} = 0$, $\exp\{jv_k\sin\gamma_0\}$, $\exp\{jv_k\sin\gamma_{M/2}\}$, $\exp\{ju_k\cos\gamma_{M/4}\}$, and $\exp\{ju_k\cos\gamma_{3M/4}\}$ become 1. Ignoring the outputs of the 0th, $M/4$th, $M/2$th, and $3M/4$th sensors, $\tilde{\boldsymbol{p}}_1(u)$, $\tilde{\boldsymbol{p}}_2(u)$, $\tilde{\boldsymbol{p}}_3(u)$ and $\tilde{\boldsymbol{p}}_4(u)$ in $\tilde{\boldsymbol{p}}(u)$ have the forms of

$$\begin{aligned} \tilde{\boldsymbol{p}}_1(u) &= [e^{ju\cos\gamma_1},\ldots,e^{ju\cos\gamma_{M/4-1}}]^T, \\ \tilde{\boldsymbol{p}}_2(u) &= [e^{ju\cos\gamma_{M/4+1}},\ldots,e^{ju\cos\gamma_{M/2-1}}]^T, \\ \tilde{\boldsymbol{p}}_3(u) &= [e^{ju\cos\gamma_{M/2+1}},\ldots,e^{ju\cos\gamma_{3M/4-1}}]^T, \\ \tilde{\boldsymbol{p}}_4(u) &= [e^{ju\cos\gamma_{3M/4+1}},\ldots,e^{ju\cos\gamma_{M-1}}]^T. \end{aligned}$$

with

$$\tilde{\boldsymbol{p}}_1(u) = \mathbf{J}\tilde{\boldsymbol{p}}_2^*(u) = \tilde{\boldsymbol{p}}_3^*(u) = \mathbf{J}\tilde{\boldsymbol{p}}_4(u).$$

As in the case of odd $M/2$, we can then follow the procedure of (18)-(21) to estimate u. However, due to the symmetry of the UCA with even number of sensors, there are repetitive terms and components with information loss in both $\tilde{\boldsymbol{p}}(u)$ and $\tilde{\boldsymbol{q}}(v)$, see (13), especially with $M/2$ being even number, which may cause degraded parameter estimation performance.

Method	Computational Complexity	Running Time ($M = 9$)	Running Time ($M = 10$)
2D-MUSIC	$O\{M^3 + (2M(M-K))N_\theta N_\phi\}$	79.03 s	79.91s
BT-ESPRIT	$O\{M^3 + 3(M-1)K^2 + 2K^3\}$	0.0091 s	0.0092 s
PE-DRC-MUSIC	$O\{D^3 + N_\theta\left[D^2(D-K) + D^2\tilde{L}_1 + \tilde{L}_1^3\right]$ $+ 2KN_\phi\left[D^2(D-K) + D^2 + D\right]\}$	14.04 s	17.82 s
RD-MUSIC	$O\{M^3(N_x+1) + MN_xK\}$ for odd sensor number $O\{M^3 + 2M^3N_x\}$ for even sensor number	0.063 s	0.071 s

TABLE III

COMPUTATIONAL COMPLEXITY AND RUNNING TIME FOR A SINGLE RUN OF 2D-MUSIC, BT-ESPRIT, PE-DRC-MUSIC AND RD-MUSIC WITH ODD AND EVEN SENSOR NUMBERS BY USING A COMPUTER EQUIPPED WITH A CPU OF 3 GHZ AND 32 GB OF RAM; $D = MN_u$ DENOTES THE DIMENSION OF THE SYNTHETIC ARRAY WITH N_u BEING THE LENGTH OF THE SAMPLING DELAY SET AND $\tilde{L}_1 = 2L_1 + 1$ IS THE TRUNCATION ORDER OF THE JACOBI–ANGER EXPANSION [26]

The DOFs of the RD-MUSIC depend on whether the number of UCA sensors is odd or even. Specifically, for odd sensor number, the DOF is $M-1$, but it reduces to $M-3$ when the UCA has an even number of sensors.

The theoretical analysis of the proposed RD-MUSIC is provided in the Appendix, and its procedure is summarized as follows:

1) Rearrange the elements of the received signal vector according to odd or even number of UCA sensors;
2) Estimate the sample covariance matrix $\mathbf{Y} = \frac{1}{N_t}\sum_{i=1}^{N_t}\boldsymbol{y}(t_i)\boldsymbol{y}^H(t_i)$, with N_t being the number of snapshots and then apply EVD on $\mathbf{Y}$ to extract the noise subspace;
3) Factorize the steering vector $\boldsymbol{a}(u,\ v)$ into two components so that u and v can be separately estimated;
4) Estimate v_k, $k = 1,\ 2, \ldots, K$, by determining the K peaks in (9) or (14);
5) Estimate u_k, $k = 1,\ 2, \ldots, K$, by finding the peak of (11) or (21);
6) With the estimated values of v_k and u_k, the DOAs of incoming signals are computed by (12).

IV. NUMERICAL EXAMPLES

In this section, 4 simulation are carried out to evaluate the performance of the proposed RD-MUSIC. A UCA with odd $M = 9$ or even number $M = 10$ sensors is considered, which is used to find the DOAs of the incoming signals. The radius of the UCA is equal to half of the wavelength of the incoming signals. There are 500 snapshots of the incoming signal observations. The following two cases are considered with different DOAs of the incoming signals:

1) Azimuth angles $\phi_1 = -10°$, $\phi_2 = 15°$; Elevation angles $\theta_1 = 20°$, $\theta_2 = 40°$;
2) Azimuth angles $\phi_1 = -5°$, $\phi_2 = 6°$; Elevation angles $\theta_1 = 40°$, $\theta_2 = 55°$.

2D-MUSIC [27], PE-DRC-MUSIC [26] and BT-ESPRIT [28] are chosen as the competing methods. In RD-MUSIC, u_k and v_k are estimated within the interval $[-\pi,\ \pi]$ with step size being 0.001π and the number of searching points being $N_x = 2000$. 2D-MUSIC requires 2D search to estimate both $\theta_k \in [0°,\ 90°]$ and $\phi_k \in [-180°,\ 180°]$ with step size being $0.1°$ and the number of searching points being $N_\theta = 900$ and $N_\phi = 3600$, respectively. PE-DRC-MUSIC constructs a virtual concentric circular array by mounting a single linear array on a rotating platform, and estimates θ and ϕ, separately. In the simulation, the search step of PE-DRC-MUSIC is identical to that of 2D-MUSIC; the linear array employs the same number of elements as the UCA, and the sampling-delay set length is $N_u = 6$. Consequently, the synthetic array comprises $D = 54$ virtual elements in the odd sensor number scenario and $D = 60$ virtual elements in the even sensor number scenario, which is 6 times the number of the applied physical UCA elements. Their computational

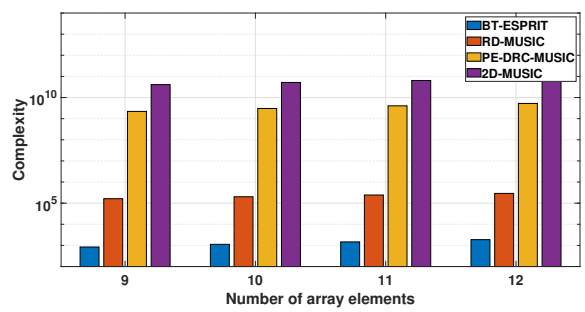


Fig. 2. Comparison of computational complexity for different methods.

complexity and running time are tabulated in Table I and Fig. 2 , respectively. BT-ESPRIT does not demand searching procedure but only DOA estimation via rotational invariance, which contributes to minimal computational loads among the 4 methods, while 2D-MUSIC and PE-DRC-MUSIC require very heavy computational complexity. RD-MUSIC aims at reducing search dimensions, and only involves 1D searches in DOA estimation, whose computational loads are much lower than that of 2D-MUSIC but higher than that of BT-ESPRIT. We can see that the average running time of 2D-MUSIC is approximately 1200 and 8700 times greater than that of RD-MUSIC and BT-ESPRIT, respectively

The first experiment is based on a single run of RD-MUSIC. SNR is fixed at 5 dB. Table II tabulates the DOA estimates for a single run of RD-MUSIC, BT-ESPRIT and 2D-MUSIC with different numbers of the UCA sensors. It can be seen that the estimated DOAs by the RD-MUSIC are closer to the true values of DOAs of the incoming signals than those of the remaining methods.

The second to third simulations provide the statistical results of the RD-MUSIC based on 200 independent trials. The root

Method \ Angle	Case I				Case II			
	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\theta}_1$	$\hat{\theta}_2$
2D-MUSIC $M = 9$	$-9.80°$	$15.20°$	$20.30°$	$40.10°$	$-4.90°$	$5.80°$	$40.30°$	$54.70°$
BT-ESPRIT $M = 9$	$-10.20°$	$15.38°$	$20.65°$	$39.88°$	$-5.28°$	$5.65°$	$40.49°$	$54.14°$
RD-MUSIC $M = 9$	$-9.91°$	$14.96°$	$19.89°$	$40.31°$	$-4.95°$	$6.15°$	$40.25°$	$55.21°$
2D-MUSIC $M = 10$	$-10.30°$	$14.80°$	$19.90°$	$39.80°$	$-4.90°$	$5.70°$	$40.50°$	$54.80°$
BT-ESPRIT $M = 10$	$-10.29°$	$15.42°$	$20.13°$	$40.21°$	$-5.26°$	$6.23°$	$40.61°$	$55.69°$
RD-MUSIC $M = 10$	$-10.24°$	$15.06°$	$20.06°$	$40.16°$	$-4.86°$	$5.87°$	$40.23°$	$54.76°$

TABLE IV
DOA ESTIMATES FOR A SINGLE RUN OF RD-MUSIC, BT-ESPRIT AND 2D-MUSIC WITH ODD AND EVEN SENSOR NUMBERS.

mean square error (RMSE) of the estimated DOA is computed as follows:

$$\text{RMSE} = \sqrt{\frac{1}{200K}\sum_{k=1}^{K}\sum_{j=1}^{200}(\hat{x}_{kj} - x_k)^2},$$

where $x_k = \theta_k$ or $x_k = \phi_k$, $\hat{x}_{kj}$ denotes either the estimated value of θ_k or ϕ_k at the jth run. The Cramér-Rao bound (CRB) is also provided as the benchmark.

In the second experiment, the performance of the RD-MUSIC versus SNR is evaluated. SNR is varied from -5 to 15 dB. Figs. 3-4 present the RMSEs on the estimated DOAs by the RD-MUSIC, 2D-MUSIC, PE-DRC-MUSIC, and BT-ESPRIT as function of SNR for odd and even numbers of UCA sensors. At low SNR, the RMSEs are large for all 4 methods, the RD-MUSIC outperforms 2D-MUSIC, PE-DRC-MUSIC, and BT-ESPRIT. In the case of both odd and even numbers of UCA sensors, the RMSEs of the RD-MUSIC and its competitors continuously decrease with the increase of SNR. The RD-MUSIC exhibits excellent DOA estimation performance, but with much smaller computational complexity. At higher SNR levels, their RMSEs converge to the same accuracy. Overall, the RD-MUSIC demonstrates superior DOA estimation performance over its competitors, with smaller RMSEs.

In the third simulation, the DOA estimation performance of the RD-MUSIC, 2D-MUSIC, PE-DRC-MUSIC, and BT-ESPRIT versus number of snapshots is examined. This test shares the same parameter settings with the second simulation. SNR is fixed at 5 dB. Figs. 5-6 plot the RMSEs versus the number of snapshots. The subspace based methods require numerous snapshots to construct the covariance matrix. The estimated covariance matrix cannot be regarded as Toeplitz matrix with small number of snapshots, which greatly impacts the DOA estimation performance. Therefore, the estimated DOAs are inaccurate, when the number of snapshots is small. Nevertheless, their RMSEs continuously decrease with the increase of the number of snapshots. As the number of snapshots grows, the enhancement in the estimation accuracy becomes less evident and their performance tends to be same. To conclude, the RD-MUSIC outperforms 2D-MUSIC, PE-DRC-MUSIC, and BT-ESPRIT for both odd and even numbers of UCA sensors.

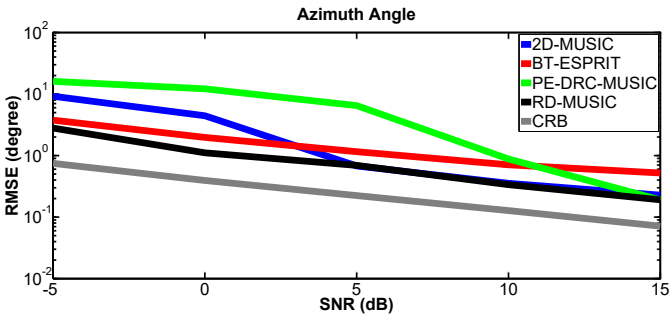
In the last simulation, to better understand the influence of angle separation, the performance of the RD-MUSIC, 2D-MUSIC, PE-DRC-MUSIC, and BT-ESPRIT versus angle separation is studied at SNR =5 dB. The combined relative RMSE (RRMSE) that incorporates both elevation and azimuth angles is calculated. Two scenarios are considered: in the first scenario, the azimuth angle of the reference signal is fixed at $\phi_1 = 0°$, and the azimuth angle of another incoming signal is $\phi_2 = \phi_1 + \Delta\phi$, with $\Delta\phi$ varying from 3° to 20°; while their elevation angles are fixed at $\theta_1 = 40°$ and $\theta_2 = 55°$; in the second scenario, the elevation angle of the reference signal is fixed at $\theta_1 = 40°$, and the elevation angle of another incoming signal is $\theta_2 = \theta_1 + \Delta\theta$, with $\Delta\theta$ varying from 3° to 20°; while their azimuth angles are fixed with $\phi_1 = -6°$ and $\phi_2 = 7°$. Other parameters are the same with that of the second experiment. As plotted in Fig. 7, all the methods fail to resolve closely spaced incident waves satisfactorily. We observe that the DOA estimation accuracy of RD-MUSIC, PE-DRC-MUSIC, BT-ESPRIT and 2D-MUSIC increases with the angle separation for both cases. Our RD-MUSIC offers the best performance with the smallest RRMSE, particularly when the angle separation is small. As the angle separation widens, the enhancement shows diminishing evidence.

V. LABORATORY AND FIELD EXPERIMENTS

To further investigate the performance of the RD-MUSIC, two experiments: a laboratory experiment and a field experiment are carried out, and their setups are shown in Figs. 8-9. Vector signal generators are used to transmit the incoming signals, which can then be captured by a UCA with 5 isotropic sensors. The UCA is employed to acquire in-phase and quadrature (IQ) data of the received signals with radius being 0.25 cm. The distance between the sources and UCA is approximately $6-10$ m to guarantee the far-field condition. In the experiment, the case of azimuthal estimation is considered at a fixed elevation angle.

Laboratory experiment employs a single 1.5 GHz signal source with 30 trials. Each run captures 2048 snapshots, ensuring temporal resolution. The true azimuth angle is 64.3°. The UCA receives the signal and feeds it to DOA estimators. BT-ESPRIT and PE-DRC-MUSIC fail in this experiment, due to insufficient number of UCA sensors. Fig. 10 shows the estimated DOAs (azimuth angles) by both MUSIC and RD-MUSIC. It is seen that the RD-MUSIC outperforms MUSIC with lower RMSE.

In the field experiment, we also conduct 30 independent trials but apply two uncorrelated signal sources operating at 1.50 GHz and 1.54 GHz, with true azimuth angles being



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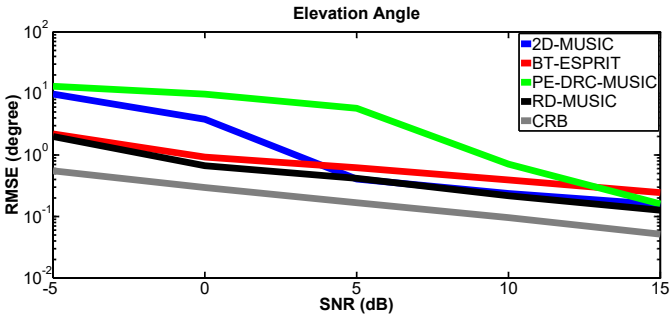

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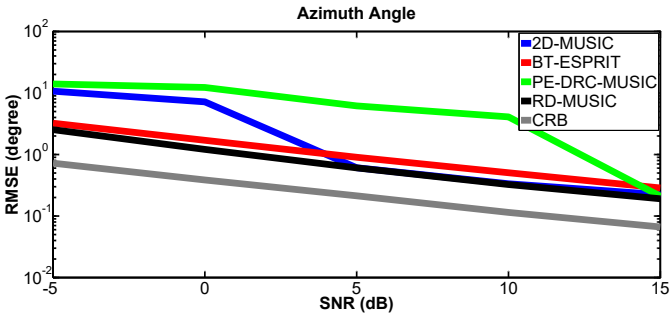

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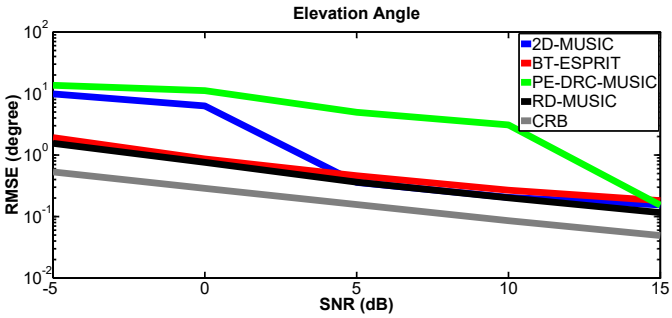
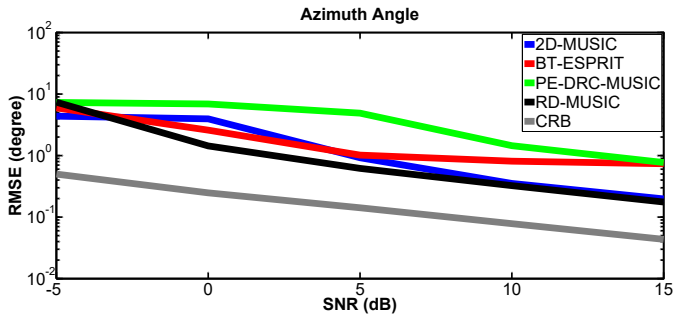


(d)

Fig. 3. Case 1: RMSEs of RD-MUSIC, BT-ESPRIT, PE-DRC-MUSIC and 2D-MUSIC versus SNR for both azimuth and elevation angles, as well as for odd and even numbers of UCA sensors: (a) azimuth angles with $M = 9$, (b) elevation angles with $M = 9$, (c) azimuth angles with $M = 10$, (d) elevation angles with $M = 10$.



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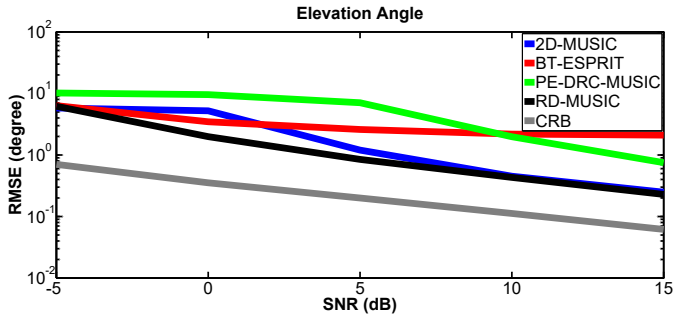

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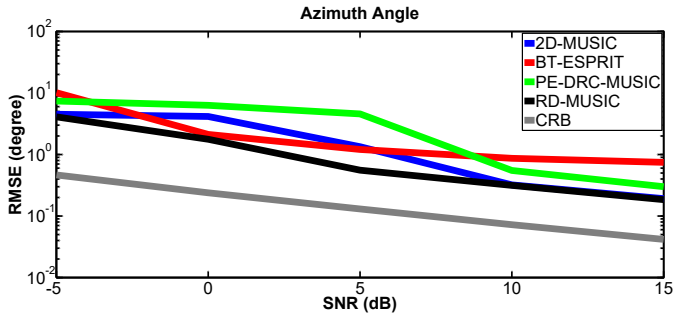

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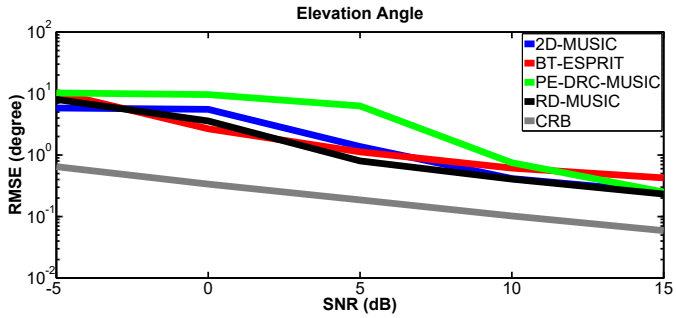

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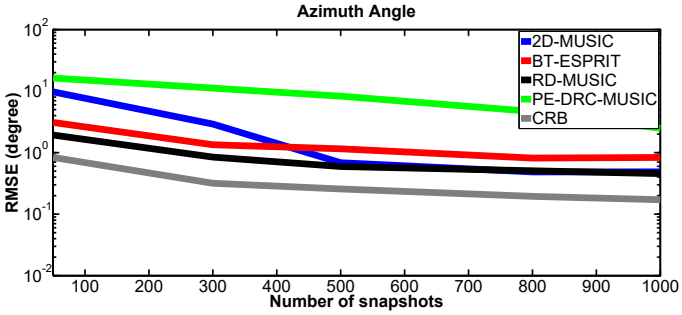
Fig. 4. Case 2: RMSEs of RD-MUSIC, BT-ESPRIT, PE-DRC-MUSIC and 2D-MUSIC versus SNR for both azimuth and elevation angles, as well as for odd and even numbers of UCA sensors: (a) azimuth angles with $M = 9$, (b) elevation angles with $M = 9$, (c) azimuth angles with $M = 10$, (d) elevation angles with $M = 10$.

0° and 98°, respectively. Similar to the laboratory test, BT-ESPRIT underperforms in the field test. As shown in Fig. 11, the DOAs (azimuth angles) of two incoming signals are accurately estimated by both MUSIC algorithm and the RD-MUSIC for these 30 runs. The proposed method provides an enhanced performance as the estimated azimuth angles are closer to the true values. The experimental results demonstrate the practical applicability and superior performance of the RD-MUSIC across both laboratory and field environments.



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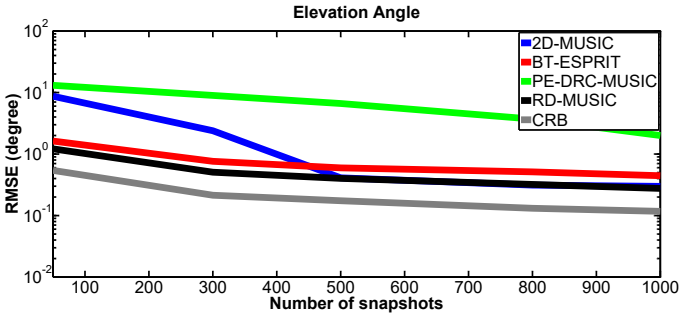

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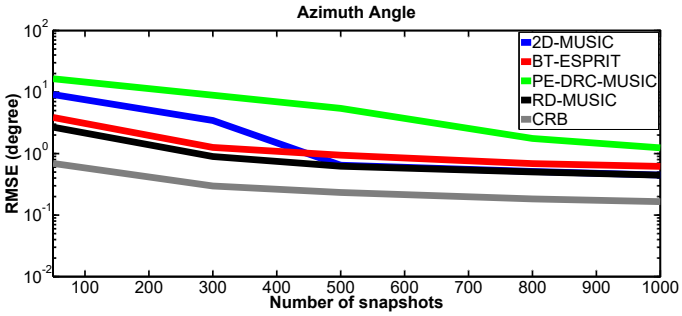

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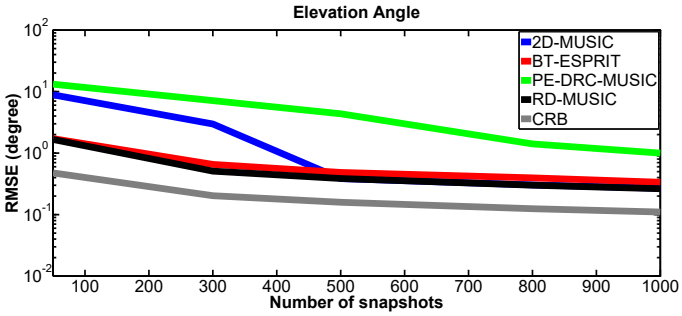

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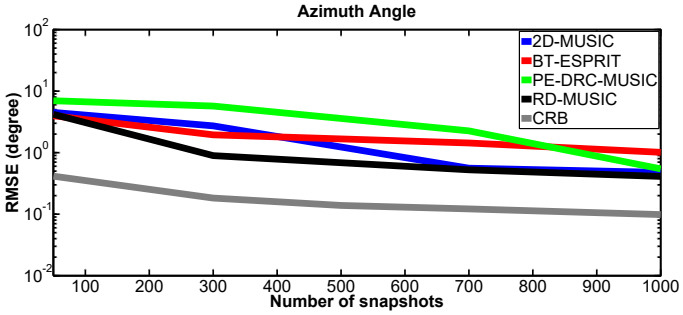
Fig. 5. Case 1: RMSEs of RD-MUSIC, BT-ESPRIT, PE-DRC-MUSIC and 2D-MUSIC versus number of snapshots for both azimuth and elevation angles, as well as for odd and even numbers of UCA sensors: (a) azimuth angles with $M = 9$, (b) elevation angles with $M = 9$, (c) azimuth angles with $M = 10$, (d) elevation angles with $M = 10$; SNR is fixed at 5 dB.



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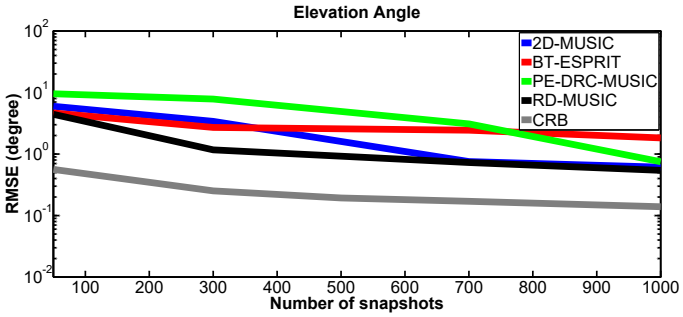

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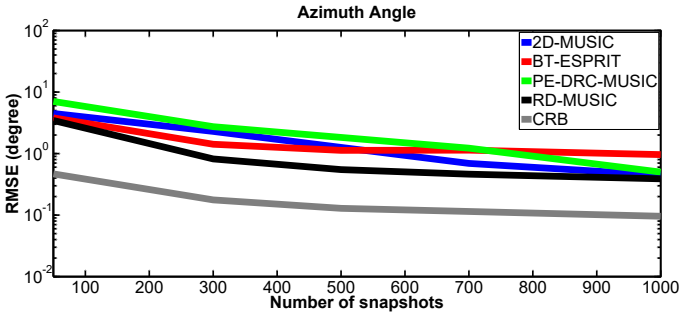

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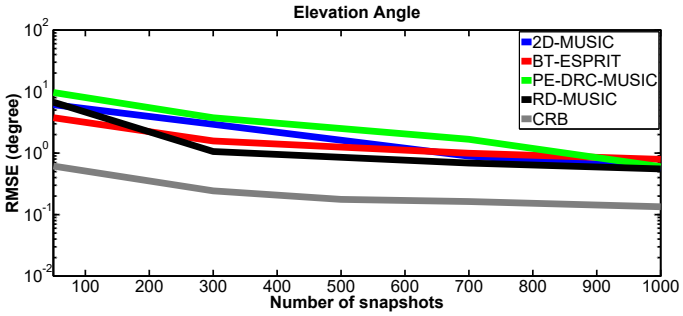

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Fig. 6. Case 2: RMSEs of RD-MUSIC, BT-ESPRIT, PE-DRC-MUSIC and 2D-MUSIC versus number of snapshots for both azimuth and elevation angles, as well as for odd and even numbers of UCA sensors: (a) azimuth angles with $M = 9$, (b) elevation angles with $M = 9$, (c) azimuth angles with $M = 10$, (d) elevation angles with $M = 10$; SNR is fixed at 5 dB.

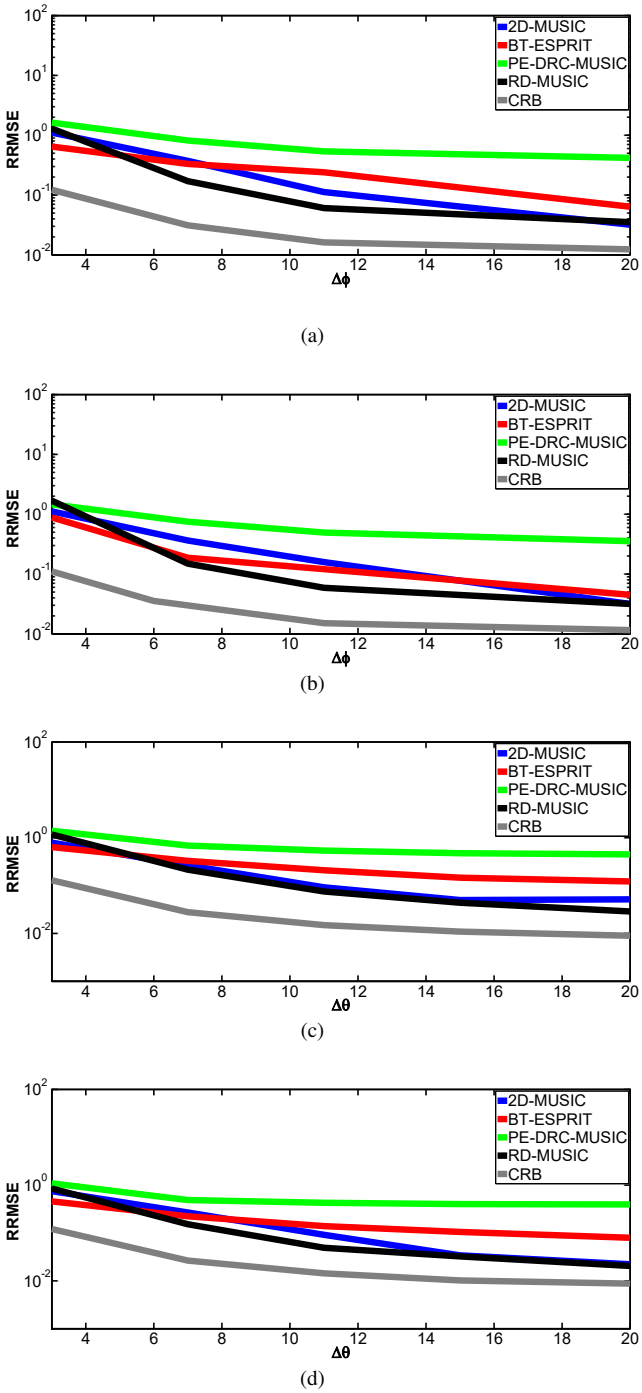

Fig. 7. RRMSEs of RD-MUSIC, BT-ESPRIT, PE-DRC-MUSIC, and 2D-MUSIC versus angle separation: (a) $M = 9$, $\phi_1 = 0°$, $\phi_2 = \phi_1 + \Delta\phi$, $\theta_1 = 40°$ and $\theta_2 = 55°$; (b) $M = 10$, $\phi_1 = 0°$, $\phi_2 = \phi_1 + \Delta\phi$, $\theta_1 = 40°$ and $\theta_2 = 55°$; (c) $M = 9$, $\theta_1 = 40°$, $\theta_2 = \theta_1 + \Delta\theta$, $\phi_1 = -6°$ and $\phi_2 = 7°$; (d) $M = 10$, $\theta_1 = 40°$, $\theta_2 = \theta_1 + \Delta\theta$, $\phi_1 = -6°$ and $\phi_2 = 7°$; SNR is fixed at 5 dB.

VI. Conclusion

This paper proposes the RD-MUSIC with UCA for DOA estimation. Instead of 2D search, we split the steering vector and apply only 1D searches in estimating the DOAs of the incoming signals for both odd and even numbers of sensors, which is much more computationally efficient. Our method provides consistent accuracy and reliability in DOA estimation

Fig. 8. Setup of the laboratory experiment.

Fig. 9. Setup of the field experiment.

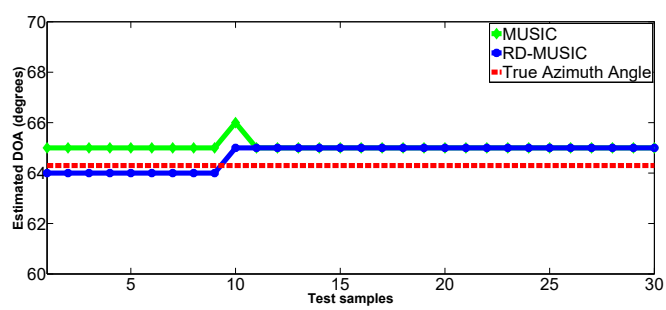

Fig. 10. DOA estimation results of laboratory experiment.

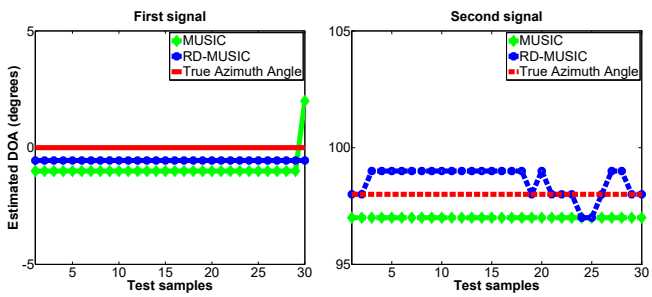

Fig. 11. DOA estimation results of field experiment.

across various UCA configurations. The RD-MUSIC is evaluated by simulated and experimental data, which demonstrates its superiority over competing methods in terms of DOA estimation accuracy and computational complexity. Future research may extend the RD-MUSIC framework to near-field scenarios, in which the steering vector is a function of both the DOA and the range [37]. By leveraging the separable structure intrinsic to the near-field steering model, we intend to develop computationally efficient algorithms capable of jointly estimating DOA and range, thereby preserving the efficiency of our methodology while achieving high-accuracy localization in near-field environments.

Appendix

The estimated values of $\hat{\theta}_k$ and $\hat{\phi}_k$ can be expressed in terms of $\hat{u}_k$ and $\hat{v}_k$ as follows:

$$\hat{\theta}_k = \theta_k + \Delta\theta_k = \arcsin\left\{\frac{\lambda\sqrt{\hat{v}_k^2+\hat{u}_k^2}}{2\pi R}\right\}$$

and

$$\hat{\phi}_k = \phi_k + \Delta\phi_k = \arctan\left\{\frac{\hat{v}_k}{\hat{u}_k}\right\}.$$

where $\hat{v}_k = v_k + \Delta v_k$ and $\hat{u}_k = u_k + \Delta u_k$ are the estimated values of v_k and u_k, respectively, with Δv_k and Δu_k being perturbations. Applying Taylor series expansion on $\hat{\theta}_k$ and $\hat{\phi}_k$ around v_k and u_k yields:

$$\hat{\theta}_k \approx \theta_k + \frac{\lambda v_k\Delta v_k + \lambda u_k\Delta u_k}{2\pi R\sqrt{(u_k^2+v_k^2)\left(1-\frac{\lambda^2}{4\pi^2R^2}(u_k^2+v_k^2)\right)}} \tag{22}$$

and

$$\hat{\phi}_k \approx \phi_k + \frac{u_k\Delta v_k - v_k\Delta u_k}{u_k^2+v_k^2}. \tag{23}$$

Therefore, the estimation bias of θ_k and ϕ_k can be expressed as

$$\begin{aligned} b_{\theta_k} &= \mathbb{E}\{\Delta\theta_k\} \\ &= \frac{\lambda v_k\mathbb{E}\{\Delta v_k\} + \lambda u_k\mathbb{E}\{\Delta u_k\}}{2\pi R\sqrt{(u_k^2+v_k^2)\left(1-\frac{\lambda^2}{4\pi^2R^2}(u_k^2+v_k^2)\right)}} \end{aligned} \tag{24}$$

and

$$b_{\phi_k} = \mathbb{E}\{\Delta\phi_k\} = \frac{u_k\mathbb{E}\{\Delta v_k\} - v_k\mathbb{E}\{\Delta u_k\}}{u_k^2+v_k^2}. \tag{25}$$

The variance of θ_k and ϕ_k estimated by RD-MUSIC can also be deduced:

$$\mathbb{E}\{\Delta\theta_k^2\} = \frac{\lambda^2 v_k^2\mathbb{E}\{\Delta v_k^2\} + \lambda^2 u_k^2\mathbb{E}\{\Delta u_k^2\}}{4\pi^2R^2(u_k^2+v_k^2)\left(1-\frac{\lambda^2}{4\pi^2R^2}(u_k^2+v_k^2)\right)} \tag{26}$$

and

$$\mathbb{E}\{\Delta\phi_k^2\} = \frac{u_k^2\mathbb{E}\{\Delta v_k^2\} + v_k^2\mathbb{E}\{\Delta u_k^2\}}{(u_k^2+v_k^2)^2}. \tag{27}$$

As the biases and variances of θ_k and ϕ_k depend on u_k and v_k, therefore, the corresponding expressions of the latter should be provided. For the odd number of sensors, we first estimate v_k, and then estimate u_k by using the former. Let $F(v)$ be the cost function of v, which can be expressed as

$$F(v) = \lambda_{\min}(v) \tag{28}$$

where $\lambda_{\min}(v)$ is the minimal eigenvalue of $\mathbf{\Psi}(v) = \mathbf{Q}^H(v)\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\mathbf{Q}(v)$. $\mathbf{\Psi}(v)$ is a Hermitian matrix, and its minimal eigenvalue can be written as

$$\lambda_{\min}(v) = \boldsymbol{q}^T\mathbf{U}_o^H(v)\mathbf{\Psi}(v)\mathbf{U}_o(v)\boldsymbol{q} = \boldsymbol{q}^T\mathbf{\Lambda}(v)\boldsymbol{q}$$

where $\mathbf{U}_o(v)$ is the non-singular matrix, $\mathbf{\Lambda}(v)$ is the diagonal matrix, whose diagonal elements are the eigenvalues of $\mathbf{\Phi}(v)$. $\boldsymbol{q}$ is an one-hot vector to extract the minimal eigenvalue $\lambda_{\min}(v)$. Define $\boldsymbol{\xi}(v) = \mathbf{Q}(v)\mathbf{U}_o(v)\boldsymbol{q}$. Therefore, $F(v) = \boldsymbol{\xi}^H(v)\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}(v)$. Apply Taylor series expansion on $F(\hat{v}_k)$ around v_k yields:

$$\begin{aligned} F(\hat{v}_k) &= F(v_k) + F'(v_k)\Delta v_k + \frac{1}{2}F''(v_k)\Delta v_k^2 + \mathcal{O}(\Delta v_k^3) \\ &\approx F(v_k) + F'(v_k)\Delta v_k + \frac{1}{2}F''(v_k)\Delta v_k^2 \end{aligned} \tag{29}$$

with

$$F'(v_k) = 2\mathrm{Re}\{\boldsymbol{\xi}^H(v_k)\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}'(v_k)\}$$

and

$$F''(v_k) = 2(\boldsymbol{\xi}'(v_k))^H\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}'(v_k)$$

where $F'(v_k)$ and $F''(v_k)$ are the first and second order derivatives of the cost function $F(v_k)$, respectively. Define $\mathbf{X} = \mathbf{A}\mathbf{S}$, with $\mathbf{S} = [\boldsymbol{s}(t_1), \ldots, \boldsymbol{s}(t_{N_t})]$. Thus,

$$\hat{\mathbf{X}} = \mathbf{X} + \tilde{\mathbf{X}} = \mathbf{X} + \mathbf{N} \tag{30}$$

where $\tilde{\mathbf{X}} = \mathbf{N} = [\boldsymbol{n}(t_1), \ldots, \boldsymbol{n}(t_{N_t})]$, is the observation noise. Applying SVD on $\mathbf{X}$ results in:

$$\mathbf{X} = \mathbf{\Sigma}_s\mathbf{\Lambda}_s\mathbf{V}_s^H. \tag{31}$$

For the finite number of snapshots, the estimated noise subspace can be written as

$$\hat{\mathbf{\Sigma}}_o = \mathbf{\Sigma}_o + \tilde{\mathbf{\Sigma}}_o \tag{32}$$

where $\tilde{\mathbf{\Sigma}}_o = -\mathbf{\Sigma}_s\mathbf{\Lambda}_s^{-1}\mathbf{V}_s^H\tilde{\mathbf{X}}^H\mathbf{\Sigma}_o$ denotes the perturbation. As $\mathbf{\Sigma}_o^H\mathbf{Q}(v_k) = \mathbf{0}$, $F'(v_k)$ and $F''(v_k)$ can be reduced to

$$\begin{aligned} F'(v_k) &\approx 2\mathrm{Re}\{\boldsymbol{\xi}^H(v_k)\tilde{\mathbf{\Sigma}}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}'(v_k)\} \\ &= -2\mathrm{Re}\{\boldsymbol{\xi}^H(v_k)\mathbf{\Sigma}_s\mathbf{\Lambda}_s^{-1}\mathbf{V}_s^H\tilde{\mathbf{X}}^H\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}'(v_k)\} \\ &= -2\mathrm{Im}\{j\boldsymbol{\xi}^H(v_k)\mathbf{\Sigma}_s\mathbf{\Lambda}_s^{-1}\mathbf{V}_s^H\tilde{\mathbf{X}}^H\mathbf{\Sigma}_o\mathbf{\Sigma}_o^H\boldsymbol{\xi}'(v_k)\} \\ &= -2\mathrm{Im}\{\boldsymbol{\alpha}_k^H\tilde{\mathbf{X}}^H\boldsymbol{\beta}_k\} \end{aligned}$$

and

$$F''(v_k) \approx 2(\boldsymbol{\xi}'(v_k))^H \boldsymbol{\Sigma}_o \boldsymbol{\Sigma}_o^H \boldsymbol{\xi}'(v_k) = 2\zeta_{vk}$$

where $\boldsymbol{\alpha}_k = \mathbf{V}_s \boldsymbol{\Lambda}_s^{-1} \boldsymbol{\Sigma}_s^H \boldsymbol{\xi}(v_k)$ and $\boldsymbol{\beta}_k = j\boldsymbol{\Sigma}_o \boldsymbol{\Sigma}_o^H \boldsymbol{\xi}'(v_k)$. As the noise is assumed to be a zero-mean white Gaussian noise with variance σ^2, we have

$$\mathbb{E}\{F'(v_k)\} = -2\text{Im}\{\boldsymbol{\alpha}_k^H \mathbb{E}\{\mathbf{N}^H\}\boldsymbol{\beta}_k\} \approx 0 \tag{33}$$

and

$$\mathbb{E}\{F'(v_k)(F'(v_k))^H\} = 4\mathbb{E}\{|\boldsymbol{\alpha}_k^H \mathbf{N}^H \boldsymbol{\beta}_k|^2\}. \tag{34}$$

The right-hand side of (34) depends solely on the perturbation matrix $\mathbf{N}$, which can be simplified to

$$\begin{aligned}\mathbb{E}\{F'(v_k)(F'(v_k))^H\} &= 4||\boldsymbol{\alpha}_k||^2 \mathbb{E}\{||\mathbf{N}||^2\}||\boldsymbol{\beta}_k||^2 \\ &= 4\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2.\end{aligned}$$

Applying Taylor series expansion on $F'(\hat{v}_k)$ around v_k:

$$F'(\hat{v}_k) = F'(v_k) + F''(v_k)\Delta v_k + \mathcal{O}(\Delta v_k^2) \tag{35}$$

Neglecting the second-order terms yield $\Delta v_k \approx -\dfrac{F'(v_t)}{F''(v_t)}$. As a results, the estimation bias of v_k by RD-MUSIC can be calculated as follows:

$$b_{vk} = \mathbb{E}\{\Delta v_k\} = -\frac{\mathbb{E}\{F'(v_k)\}}{F''(v_k)} = 0 \tag{36}$$

and the variance of v_k by RD-MUSIC can also de developed:

$$\begin{aligned}\mathbb{E}\{\Delta v_k^2\} = \text{MSE}_{vk} &= \frac{\mathbb{E}\{F'(v_k)(F'(v_k))^H\}}{(F''(v_k))^2} \\ &= \frac{\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2}{\zeta_{vk}}.\end{aligned} \tag{37}$$

With estimated value $\hat{v}_k$, the cost function $H(u)$ is employed to estimate u_k.

$$H(u) = \boldsymbol{p}^H(u)\boldsymbol{\Psi}(\hat{v}_k)\boldsymbol{p}(u) \tag{38}$$

Applying Taylor series expansion on $H(\hat{u}_k)$ around u_k:

$$\begin{aligned}H(\hat{u}_k) &= H(u_k) + H'(u_k)\Delta u_k + \frac{1}{2}H''(u_k)\Delta u_k^2 + \mathcal{O}(\Delta u_k^3) \\ &\approx H(u_k) + H'(u_k)\Delta u_k + \frac{1}{2}H''(u_k)\Delta u_k^2\end{aligned}$$

with

$$H'(u_k) = 2\text{Re}\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}(\hat{v}_k)\boldsymbol{p}'(u_k)\}$$

and

$$H''(u_k) = 2(\boldsymbol{p}'(u_k))^H \boldsymbol{\Psi}(\hat{v}_k)\boldsymbol{p}'(u_k) = \zeta_{uk}$$

where $H'(u_k)$ and $H''(u_k)$ are the first and second order derivatives of the cost function $H(u_k)$, respectively. Applying Taylor series expansion on $\boldsymbol{\Psi}(\hat{v}_k)$ around v_k:

$$\boldsymbol{\Psi}(\hat{v}_k) \approx \boldsymbol{\Psi}(v_k) + \boldsymbol{\Psi}'(v_k)\Delta v_k + \frac{1}{2}\boldsymbol{\Psi}''(v_k)\Delta v_k^2 \tag{39}$$

As $\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}(v_k) = \mathbf{0}$, $H'(u_k)$ can be reduced to

$$H'(u_k) = 2\text{Re}\{\boldsymbol{p}^H(u_k)\{\boldsymbol{\Psi}'(v_k)\Delta v_k + \frac{1}{2}\boldsymbol{\Psi}''(v_k)\Delta v_k^2\}\boldsymbol{p}'(u_k)\}.$$

Then, the expressions of $\mathbb{E}\{H'(u_k)\}$ and $\mathbb{E}\{H'(u_k)(H'(u_k))^H\}$ can be deduced as follows:

$$\begin{aligned}\mathbb{E}\{H'(u_k)\} &= 2\text{Re}\{\boldsymbol{p}^H(u_k)\mathbb{E}\{\boldsymbol{\Psi}'(v_k)\Delta v_k \\ &+ \frac{1}{2}\boldsymbol{\Psi}''(v_k)\Delta v_k^2\}\boldsymbol{p}'(u_k)\} \\ &= 2\text{Re}\{\boldsymbol{p}^H(u_k)\{\boldsymbol{\Psi}'(v_k)\mathbb{E}\{\Delta v_k\} \\ &+ \frac{1}{2}\boldsymbol{\Psi}''(v_k)\mathbb{E}\{\Delta v_k^2\}\}\boldsymbol{p}'(u_k)\} \\ &= \text{Re}\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}''(v_k)\mathbb{E}\{\Delta v_k^2\}\boldsymbol{p}'(u_k)\} \\ &= \sigma^2\frac{||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2}{\zeta_{vk}}\text{Re}\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}''(v_k)\boldsymbol{p}'(u_k)\} \\ &= \frac{\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2\gamma_{uk}}{\zeta_{vk}}\end{aligned}$$

and

$$\begin{aligned}&\mathbb{E}\{H'(u_k)(H'(u_k))^H\} = 4\text{Re}\Big\{\boldsymbol{p}^H(u_k)\mathbb{E}\Big\{\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}'(v_k))^H\Delta v_k^2 \\ &+ \frac{1}{4}\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}''(v_k))^H\Delta v_k^4 + \frac{1}{2}\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}''(v_k))^H\Delta v_k^3 \\ &+ \frac{1}{2}\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}'(v_k))^H\Delta v_k^3\Big\}\boldsymbol{p}(u_k)\Big\} \\ &= 4\text{Re}\Big\{\boldsymbol{p}^H(u_k)\mathbb{E}\Big\{\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}'(v_k))^H\Delta v_k^2 \\ &+ \frac{1}{4}\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}''(v_k))^H\Delta v_k^4\Big\}\boldsymbol{p}(u_k)\Big\} \\ &= 4\text{Re}\Big\{\boldsymbol{p}^H(u_k)\Big\{\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}'(v_k))^H\frac{\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2}{\zeta_{vk}} \\ &+ \frac{1}{4}\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}''(v_k))^H\frac{\sigma^4||\boldsymbol{\alpha}_k||^4||\boldsymbol{\beta}_k||^4}{\zeta_{vk}^2}\Big\}\boldsymbol{p}(u_k)\Big\} \\ &\frac{4\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2}{\zeta_{vk}}\text{Re}\Big\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}'(v_k))^H\boldsymbol{p}(u_k)\Big\} \\ &+ \frac{\sigma^4||\boldsymbol{\alpha}_k||^4||\boldsymbol{\beta}_k||^4}{\zeta_{vk}^2}\text{Re}\Big\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}''(v_k))^H\boldsymbol{p}(u_k)\Big\} \\ &= \frac{4\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2}{\zeta_{vk}}\rho_{uk1} + \frac{\sigma^4||\boldsymbol{\alpha}_k||^4||\boldsymbol{\beta}_k||^4}{\zeta_{vk}^2}\rho_{uk2}\end{aligned}$$

with

$$\rho_{uk1} = \text{Re}\Big\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}'(v_k)(\boldsymbol{\Psi}'(v_k))^H\boldsymbol{p}(u_k)\Big\}$$

and

$$\rho_{uk2} = \text{Re}\Big\{\boldsymbol{p}^H(u_k)\boldsymbol{\Psi}''(v_k)(\boldsymbol{\Psi}''(v_k))^H\boldsymbol{p}(u_k)\Big\}$$

Therefore, the estimation bias and variance of u_k by RD-MUSIC are

$$b_{uk} = \mathbb{E}\{\Delta u_k\} = -\frac{\mathbb{E}\{H'(v_k)\}}{H''(v_k)} = -\frac{\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2\gamma_{uk}}{\zeta_{vk}\zeta_{uk}} \tag{40}$$

and

$$\begin{aligned}\mathbb{E}\{\Delta u_k^2\} &= \frac{\mathbb{E}\{H'(u_k)(H'(u_k))^H\}}{(H''(u_k))^2} \\ &= \frac{4\sigma^2||\boldsymbol{\alpha}_k||^2||\boldsymbol{\beta}_k||^2\rho_{uk1}}{\zeta_{vk}\zeta_{uk}^2} + \frac{\sigma^4||\boldsymbol{\alpha}_k||^4||\boldsymbol{\beta}_k||^4\rho_{uk2}}{\zeta_{vk}^2\zeta_{uk}^2}\end{aligned} \tag{41}$$

and the mean square error of u_k is then:

$$\text{MSE}_{uk} = b_{uk}^2 + \mathbb{E}\{\Delta u_k^2\}. \tag{42}$$

For the even number of sensors, v_k and u_k are estimated by (15) and (21), separately. Their cost functions share same structure with that of (28). It can be easily deduced that the RD-MUSIC is an unbiased estimator for both u_k and v_k in the scenario of even sensor number. The variances of both u_k and v_k have similar expression with that of (37), which depend on σ^2. With the estimation biases and variances for u_k and v_k, the DOA estimation accuracy can then be obtained.

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